

Linear equations: LU decomposition

Consider solving a linear system of equations, say with 3 variables

$$\mathbf{A} \cdot \mathbf{x} = \mathbf{b} \rightarrow \mathbf{A} = \mathbf{L} \cdot \mathbf{U}$$

Matrix $\mathbf{A}$ is factorized or **decomposed** into a product of *lower triangular* $\mathbf{L}$ and *upper triangular* $\mathbf{U}$ matrices,

$$\mathbf{L} = \begin{pmatrix} \ell_{00} & 0 & 0 \\ \ell_{10} & \ell_{11} & 0 \\ \ell_{20} & \ell_{21} & \ell_{22} \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} u_{00} & u_{01} & u_{02} \\ 0 & u_{11} & u_{12} \\ 0 & 0 & u_{22} \end{pmatrix}$$

Looks formidable but very useful and not nearly as hard.
Multiplying

$$\begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \ell_{00}u_{00} & \ell_{00}u_{01} & \ell_{00}u_{02} \\ \ell_{10}u_{00} & \ell_{10}u_{01} + \ell_{11}u_{11} & \ell_{10}u_{02} + \ell_{11}u_{12} \\ \ell_{20}u_{00} & \ell_{20}u_{01} + \ell_{21}u_{11} & \ell_{20}u_{02} + \ell_{21}u_{12} + \ell_{22}u_{22} \end{pmatrix}$$

But without proper ordering, the factorization may fail!

LU Decomposition

The matrix multiplication $\mathbf{L} \cdot \mathbf{U}$ yields

$$\begin{array}{lll} \ell_{00}u_{00} = a_{00} & \ell_{00}u_{01} = a_{01} & \ell_{00}u_{02} = a_{02} \\ \ell_{10}u_{00} = a_{10} & \ell_{10}u_{01} + \ell_{11}u_{11} = a_{11} & \ell_{10}u_{02} + \ell_{11}u_{12} = a_{12} \\ \ell_{20}u_{00} = a_{20} & \ell_{20}u_{01} + \ell_{21}u_{11} = a_{21} & \ell_{20}u_{02} + \ell_{21}u_{12} + \ell_{22}u_{22} = a_{22} \end{array}$$

A catch : $3 \times 3 = 9$ equations but $3 \times (3 + 1)$ variables!!

Trick : Either all three $\ell_{ij} = 1$, called *Doolittle* or all three $u_{ij} = 1$, called *Crout* decomposition.

Any of the decomposition of $\mathbf{L}$ and $\mathbf{U}$ can proceed iteratively.

Doolittle : $\ell_{00} = \ell_{11} = \ell_{22} = 1$ and *Crout* : $u_{00} = u_{11} = u_{22} = 1$

This straight away implies

$$\begin{array}{ll} \text{Doolittle :} & u_{00} = a_{00}, u_{01} = a_{01}, u_{02} = a_{02} \\ \text{Crout :} & \ell_{00} = a_{00}, \ell_{10} = a_{10}, \ell_{20} = a_{20} \end{array}$$

Rest of ℓ_{ij} / u_{ij} can be solved from remaining equations to achieve $\mathbf{LU}$.

Doolittle LU

1. Remember $u_{00} = a_{00}$, $u_{01} = a_{01}$ and $u_{02} = a_{02} \Rightarrow u_{0j} = a_{0j}$
2. Do the calculation in the order they appear

$u_{00} = a_{00}$	$\ell_{00} = 1$
$u_{01} = a_{01}$	$\ell_{01} = 0$
$u_{02} = a_{02}$	$\ell_{02} = 0$
$u_{10} = 0,$	$\ell_{10} = (a_{10})/u_{00}$
$u_{20} = 0,$	$\ell_{20} = (a_{20})/u_{00}$
$u_{11} = a_{11} - \ell_{10}u_{01},$	$\ell_{11} = 1$
$u_{21} = 0,$	$\ell_{21} = (a_{21} - \ell_{20}u_{01})/u_{11}$
$u_{12} = a_{12} - \ell_{10}u_{02},$	$\ell_{12} = 0$
$u_{22} = a_{22} - \ell_{20}u_{02} - \ell_{21}u_{12},$	$\ell_{22} = 1$

Generic form for each $j = 0, 1, \dots, N-1$, in order they appear, for $i = 1, 2, \dots, N-1$

$$u_{ij} = a_{ij} - \sum_{k=0}^{i-1} \ell_{ik} u_{kj} \quad \text{for } i = 2, \dots, j \quad \text{i.e. } i \leq j$$

$$\ell_{ij} = \left(a_{ij} - \sum_{k=0}^{j-1} \ell_{ik} u_{kj} \right) / u_{jj} \quad \text{for } i = j+1, j+2, \dots, N-1 \quad \text{i.e. } i > j$$

LU storage

The above can be coded far more efficiently by directly using a_{ij} where ever u_{ij} or l_{ij} appears. Why?

Important : Every a_{ij} is used only once and never again.

$\Rightarrow u_{ij}, l_{ij}$ can be stored in the same location / memory of a_{ij} .
Hence memory requirement is half.

Storing Doolittle LU decomposed matrix

$$\mathbf{LU} = \begin{pmatrix} u_{00} & u_{01} & u_{02} \\ \ell_{10} & u_{11} & u_{12} \\ \ell_{20} & \ell_{21} & u_{22} \end{pmatrix} \rightarrow \mathbf{A}$$

No need to store $\ell_{ii} = 1$, modify loop over indices accordingly.

Otherwise, numerical cost of Gauss-Jordan and LU are same,
 $\mathcal{O}(N^3)$.

But, can LU be always done?

Can we always LU decompose?

- If $a_{00} = 0$, then either **L** or **U** is singular $\rightarrow$ impossible if **A** is not.

Solution : Row pivot

- Guaranteed if all *leading submatrices* have nonzero determinant

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \Rightarrow \mathbf{A}_1 = 1, \mathbf{A}_2 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \text{ and } \mathbf{A}_3 = \begin{pmatrix} 1 & 0 & 2 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\det[\mathbf{A}_1] = 1, \det[\mathbf{A}_2] = 1 \text{ and } \det[\mathbf{A}_3] = -3$$

- Additional pivoting if determinant of any leading submatrix is zero but the matrix itself is invertible.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 0 & 2 \\ -1 & 1 & 2 \end{pmatrix} \Rightarrow \mathbf{A}_1 = 1, \mathbf{A}_2 = \begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix} \text{ and } \mathbf{A}_3 = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 0 & 2 \\ -1 & 1 & 2 \end{pmatrix}$$

$$\det[\mathbf{A}_1] = 1, \det[\mathbf{A}_2] = 0 \text{ and } \det[\mathbf{A}_3] = 4$$

- Otherwise, no solution exists and your are doomed!

U in Gauss-Jordan

Recall the **U** matrix needed for determinant calculation in Gauss-Jordan elimination

$$\begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix} \Rightarrow \begin{pmatrix} u_{00} & u_{01} & u_{02} \\ 0 & u_{11} & u_{12} \\ 0 & 0 & u_{22} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \bar{b}_0 \\ \bar{b}_1 \\ \bar{b}_2 \end{pmatrix}$$

Solutions x_i starts from last row i.e. by *backward substitution*

$$u_{22}x_2 = \bar{b}_2 \quad x_2 = \frac{\bar{b}_2}{u_{22}}$$

$$u_{11}x_1 + u_{12}x_2 = \bar{b}_1 \quad x_1 = \frac{\bar{b}_1 - u_{12}x_2}{u_{11}}$$

$$u_{00}x_0 + u_{01}x_1 + u_{02}x_2 = \bar{b}_0 \quad x_0 = \frac{\bar{b}_0 - u_{01}x_1 - u_{02}x_2}{u_{00}}$$

A generic solution for $N \times N$ matrix by backward substitution is

$$x_i = \frac{1}{u_{ii}} \left(\bar{b}_i - \sum_{j=i+1}^N u_{ij}x_j \right), \text{ where } x_N = \frac{\bar{b}_N}{u_{NN}} \text{ and } i = N-2, N-3, \dots, 0$$

LU forward-backward

To solve linear system of equations using **LU** decomposition it is advisable to begin with **partial pivoting**.

★ In the next step, consider the following split up

$$\mathbf{A} \cdot \mathbf{x} = \mathbf{b} \Rightarrow \mathbf{L} \cdot (\mathbf{U} \cdot \mathbf{x}) = \mathbf{b} \quad \Bigg| \quad \mathbf{U} \cdot \mathbf{x} = \mathbf{y} \Rightarrow \mathbf{L} \cdot \mathbf{y} = \mathbf{b}$$

★ First solve for **y** from $\mathbf{L} \cdot \mathbf{y} = \mathbf{b}$ using *forward substitution*, then use it to solve for **x** by *backward substitution* from $\mathbf{U} \cdot \mathbf{x} = \mathbf{y}$

Forward substitution :

$$\begin{pmatrix} 1 & 0 & 0 \\ \ell_{01} & 1 & 0 \\ \ell_{20} & \ell_{21} & 1 \end{pmatrix} \begin{pmatrix} y_0 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix}$$

Starting from the **first row** i.e. moving **forward** we solve for y_i

$$y_0 = b_0$$

$$\ell_{10}y_0 + y_1 = b_1$$

$$\ell_{20}y_0 + \ell_{21}y_1 + y_2 = b_2$$

$$y_0 = b_0$$

$$y_1 = b_1 - \ell_{10}y_0$$

$$y_2 = b_2 - \ell_{20}y_0 - \ell_{21}y_1$$

Backward substitution :

$$\begin{pmatrix} u_{00} & u_{01} & u_{02} \\ 0 & u_{11} & u_{12} \\ 0 & 0 & u_{22} \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ y_2 \end{pmatrix}$$

Starting from the **last row** i.e. moving **backward** to solve for x_i

$$u_{22}x_2 = y_2 \quad x_2 = y_2 / u_{22}$$

$$u_{11}x_1 + u_{12}x_2 = y_1 \quad x_1 = (y_1 - u_{12}x_2) / u_{11}$$

$$u_{00}x_0 + u_{01}x_1 + u_{02}x_2 = y_0 \quad x_0 = (y_0 - u_{01}x_1 - u_{02}x_2) / u_{00}$$

In generic form, the solutions for y_i and subsequently x_i are

$$y_i = b_i - \sum_{j=0}^{i-1} \ell_{ij}y_j, \quad \text{where } y_0 = b_0 \quad \text{and } i = 1, 2, \dots, N-1$$

$$x_i = \frac{1}{u_{ii}} \left(y_i - \sum_{j=i+1}^N u_{ij}x_j \right), \quad \text{where } x_{N-1} = \frac{y_{N-1}}{u_{N-1,N-1}} \quad \text{and } i = N-2, N-3, \dots, 0$$

★ Get determinant of **A** for free

$$\det \mathbf{A} = \det \mathbf{LU} = \det \mathbf{L} \times \det \mathbf{U} = (-1)^n \prod_i u_{ii}$$

★ For inverse, iterate through each column of the identity matrix.

Cholesky decomposition

Properties and symmetries of **A** are often used to simplify the process of solving linear equations.

Cholesky decomposition : factorization of **Hermitian, positive definite** matrix (which often is the case in physics e.g. covariance matrix) into a product of **L** and **L^T**.

For a 3×3 system,

$$\mathbf{A} = \mathbf{L}\mathbf{L}^\dagger \xrightarrow{\text{real}} \mathbf{L}\mathbf{L}^T \Rightarrow \begin{pmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} \ell_{00} & 0 & 0 \\ \ell_{10} & \ell_{11} & 0 \\ \ell_{20} & \ell_{21} & \ell_{22} \end{pmatrix} \begin{pmatrix} \ell_{00} & \ell_{01} & \ell_{02} \\ 0 & \ell_{11} & \ell_{12} \\ 0 & 0 & \ell_{22} \end{pmatrix}$$

where $\ell_{ij} = \ell_{ji}$. When **A** is real and positive definite,

$$\ell_{ii} = \pm \sqrt{a_{ii} - \sum_{j=0}^{i-1} \ell_{ij}^2} \quad \text{and} \quad \ell_{ij} = \frac{1}{\ell_{ii}} \left(a_{ij} - \sum_{k=0}^{i-1} \ell_{ik} \ell_{kj} \right) \quad \text{for } j > i$$

Signs before square roots are inconsequential. **DIY** the decomposition for complex matrix.

Cholesky

Cholesky is about **TWICE** as efficient as the LU decomposition for solving system of linear equations.

An example of Cholesky decomposition of a real, symmetric matrix is (taken from Wikipedia),

$$\begin{pmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 6 & 1 & 0 \\ -8 & 5 & 3 \end{pmatrix} \begin{pmatrix} 2 & 6 & -8 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{pmatrix}$$

Apart from being used for numerical solution of linear equations, Cholesky decomposition is also used in non-linear optimization for multiple variable, monte carlo simulation for decomposing covariance matrix, inversion of Hermitian matrices etc.